

DentaVision: Automating Intraoral Scanning & Digital 3D Dental Modeling with Robotics

Marvin Kraus Sebastian Nagles Irmak Damla Özdemir Manuel Stöth
Pascal Meißner

{marvin.kraus | sebastian.nagles | irmakdamla.oezdemir | manuel.stoeth}@study.thws.de
{pascal.meissner}@thws.de

Faculty of Computer Science and Business Information Systems (FIW)
Technical University of Applied Sciences Wuerzburg-Schweinfurt (THWS)
Sanderheinrichsleitenweg 20, 97074 Würzburg, Germany

Abstract—Intraoral scanning is commonly performed manually by trained dental personnel, making the process costly, time-consuming, and dependent on operator experience. At the same time, automating this task is challenging because a robot must move safely around complex dental geometry while capturing sufficient data without causing any harm to the patient. This project investigates whether existing robotics, scanning, and computer vision technologies can be combined into an automated system for intraoral dental scanning and modeling.

The project aims to be able to automate the process in an environment, where the positions of the robot arm and an inert teeth model, in the form of an artificial head, are precisely known and don't change. The concrete setup is roughly as follows: A dental scanner is mounted onto a robotic arm using a custom 3D-printed housing, which allows controlled movement of the scanner around the previously mentioned teeth model. This is then operated through a multitude of preexisting components and libraries, that are chained together in order to let the robot autonomously move around the teeth model. The basic concept that emanates from that is a loop, in which the robot moves to a position, takes a scan, and with that updates the current digital model of the teeth. The updated digital model is then used to derive the next scanning position. This repeats until the digital model reaches a certain level of accuracy, which is detailed further below.

The scope of the project includes the mechanical integration of the scanner, the generation of suitable robot trajectories, the acquisition of image or scan data, and the pipeline of repeatedly modeling the scan data into a digital model of the teeth. The target pose estimation for the next best scan, which is needed for this project, will be contributed by an external student as part of their bachelor's thesis.

Evaluation of the system is realized by analyzing scan coverage and registration quality of the generated digital model. This means that certain regions of the teeth have to be completely modeled and, to a degree, shouldn't have any gaps in them. The resulting model is then also compared with the manually performed scan.

The resulting system may provide a basis for further investigating automated intraoral scanning under controlled conditions. In the future, the results of this project may be used to create a specialized medical robot, designed to be certifiable to work with real humans, thus making trained personnel available for other tasks.

Index Terms—UR5e, Simulation, etc

I. SIMULATION-BASED IMAGE GENERATION

Authors: Manuel Stöth

A. Introduction

II. TITLE

Authors: Sebastian Nagles

A. Introduction

III. TITLE

Authors: Marvin Kraus

A. Introduction

IV. TITLE

Authors: Irmak Damla Özdemir

A. Introduction

V. CONCLUSION

LIST OF FIGURES